BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 123 · Lesson 123 · Programming Paradigms: How We Express Solutions

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What development approach is most explicitly risk-driven?

Q2. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q3. What is the time complexity of sequential search in the worst case?

Q4. After pushing A then B onto a stack, which is popped first?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

paradigm — Style or model for expressing computations.

imperative — Programming by explicit state-changing commands.

declarative — Programming by describing desired result/relations.

object-oriented — Programming using interacting objects with state and behaviour.

inheritance — A derived class reuses and specialises a base class.

override — A derived class replaces an inherited virtual method implementation.

polymorphism — One base-class method call can run different implementations according to the actual object type.

New learning

Imperative/procedural — Express a sequence of state-changing instructions and reusable procedures.

Declarative — Describe a desired result or logical relationship rather than an explicit control-flow recipe.

Object-oriented — Objects group state and behaviour. Inheritance lets a derived class reuse a base class; an override replaces a virtual method. Calling that method through a base-class reference runs the actual object’s version: polymorphism .

Worked example

Compare a SQL query selecting all orders over £100 with a loop that scans each record.

1. Model answer — SQL describes the required set of matching records (declarative); the loop specifies iteration, testing and append operations (imperative/procedural). Both can produce equivalent output through different expression styles. — Check: link each claim to the scenario and use precise OCR terminology.

2. Find the parent and children — Alert is the base class; EmailAlert : Alert and PopupAlert : Alert are derived classes. Each is a kind of Alert and can be stored in an Alert[] .

3. Same signature, specialised behaviour — Show() is virtual in Alert . Both derived classes use override to supply their own implementation.

4. Trace the output — a.Show() prints Email then Popup . Although a is declared as Alert , the runtime selects the implementation belonging to the actual object. This is polymorphism, not two separate if tests.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

We Do · Practise with guidance

A simulation models each vehicle with position, speed and update behaviour. Which paradigm may fit?

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Object-oriented design can group vehicle state and operations within objects. A procedural implementation is possible too; justify selection by encapsulation and reuse rather than claiming one paradigm is always best.

Which Speak() method runs?

```
using System;
Animal[] pets = { new Cat(), new Dog() };
foreach (Animal pet in pets)
    Console.WriteLine(pet.Speak());

class Animal {
    public virtual string Speak() => "...";
}
class Dog : Animal {
    public override string Speak() => "Woof";
}
class Cat : Animal {
    public override string Speak() => "Miaow";
}
```

Predict the output first

Name the base class and two derived classes. In which order are the actual objects stored in the array?

Apply dynamic dispatch

The first object is a Cat , so its overridden Speak() runs. The second is a Dog . No type-checking if statement is needed.

Explain the actual result

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Outputs Miaow then Woof . Animal is the declared reference type; each object’s runtime type determines the overridden method.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define a programming paradigm and contrast imperative with declarative languages. [6]

Question 2

Explain how a SQL query differs conceptually from a loop-based filter. [6]

Question 3

A base class Alert declares virtual Show(). Two derived classes EmailAlert and PopupAlert override Show(). An Alert[] contains one of each. Explain inheritance, overriding, polymorphism and one reason this design may be preferable to procedural type-checking. [8]

Exit Ticket · Answer before checking

Declarative programming always requires the programmer to specify every loop and update.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Spiral.

Q2. X is removed; Y remains at the front.

Q3. O(n).

Q4. B, because a stack is LIFO.

You Do pages · Independent questions

Q1. Paradigm is an organising programming approach [2]; imperative how/state changes [2]; declarative required result/relations [2].

Q2. SQL expresses target set/condition [3]; procedural loop controls traversal and selection explicitly [3].

Q3. Inheritance reuses/specialises Alert [2]; overriding supplies each derived class's Show implementation [2]; a call through Alert uses the object's actual type at runtime [2]; shared base contract supports extension without scattered type-checking branches [2].

Exit Ticket · Model answer

False. It emphasises the result or relation rather than control flow.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.